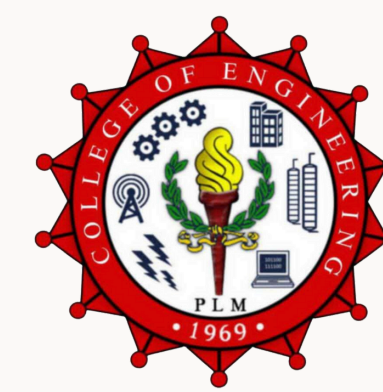




SANAOil

Computer Vision-Based Watercraft for Oil Spill Assistance Using YOLOv8 and Physical Recovery with Belt Skimming



INTRODUCTION

The Philippines, as an archipelagic nation, is vulnerable to oil spills that damage ecosystems, disrupt livelihoods, and pose health risks. Manual cleanup methods are risky and often ineffective. SANAOil offers an autonomous, vision-based solution for faster, safer oil spill response.

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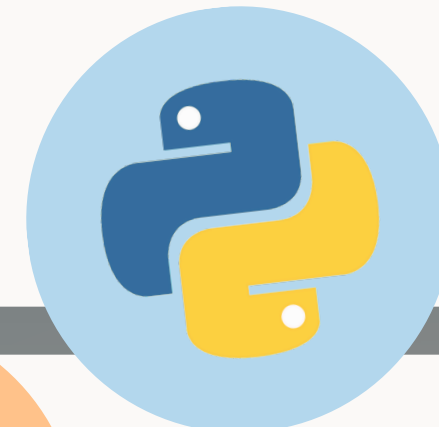
Oil Spill Incidents in the Philippines from 2000 to 2022

METHODS



Detection System

YOLOv8 + ESP32-CAM for real-time oil identification



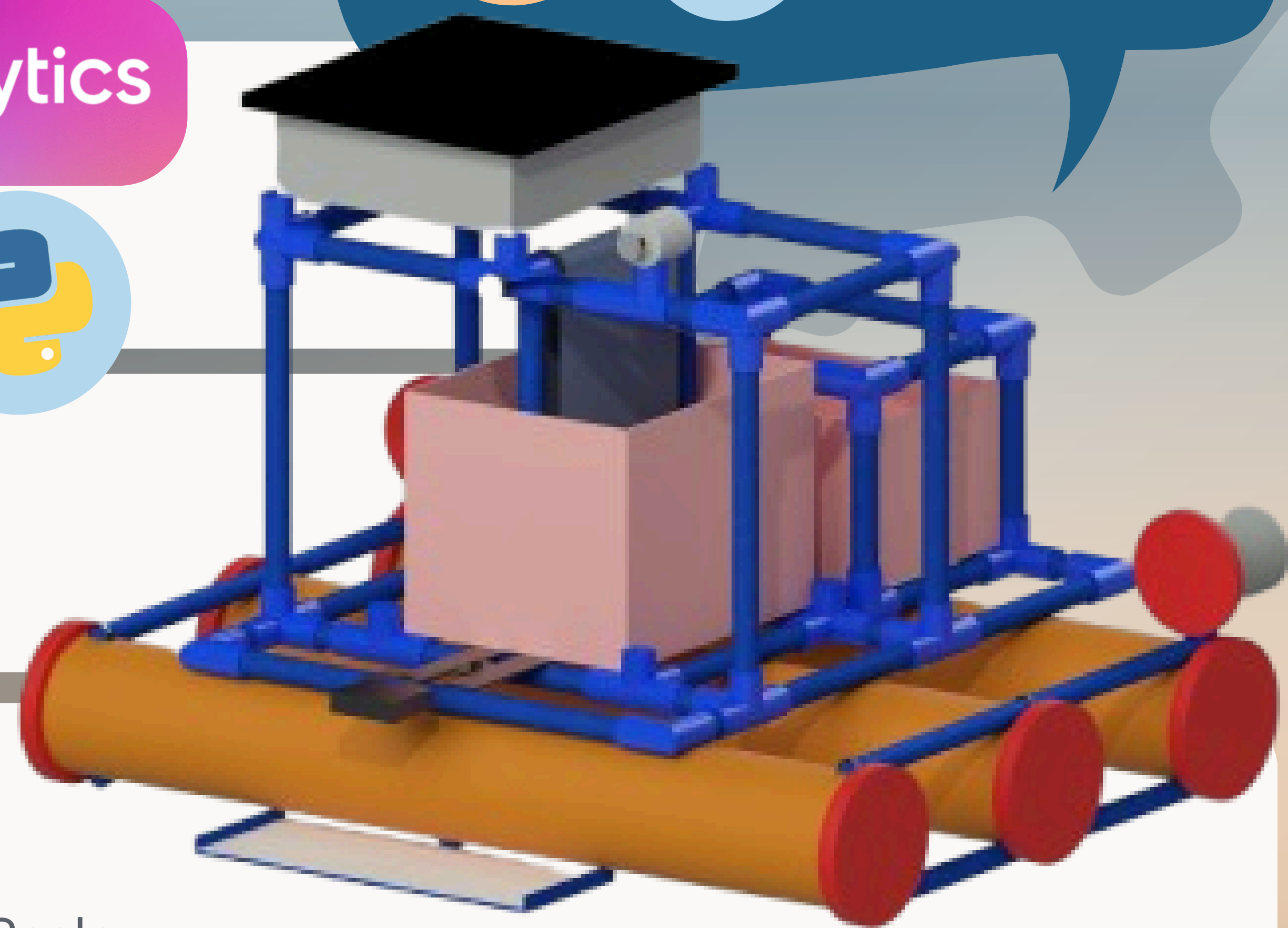
Mechanical Recovery

Watercraft with pump + belt skimmer system



Watercraft Testing

- Field deployment in Laguna (river, lake)
- Scaled simulations using Douglas Sea Scale
- ISO-compliant safety and system checks



SDG

14

Life Below Water

13

Climate Action

8

Decent Work and Economic Growth

1

No Poverty



RESULTS & DISCUSSION

90%

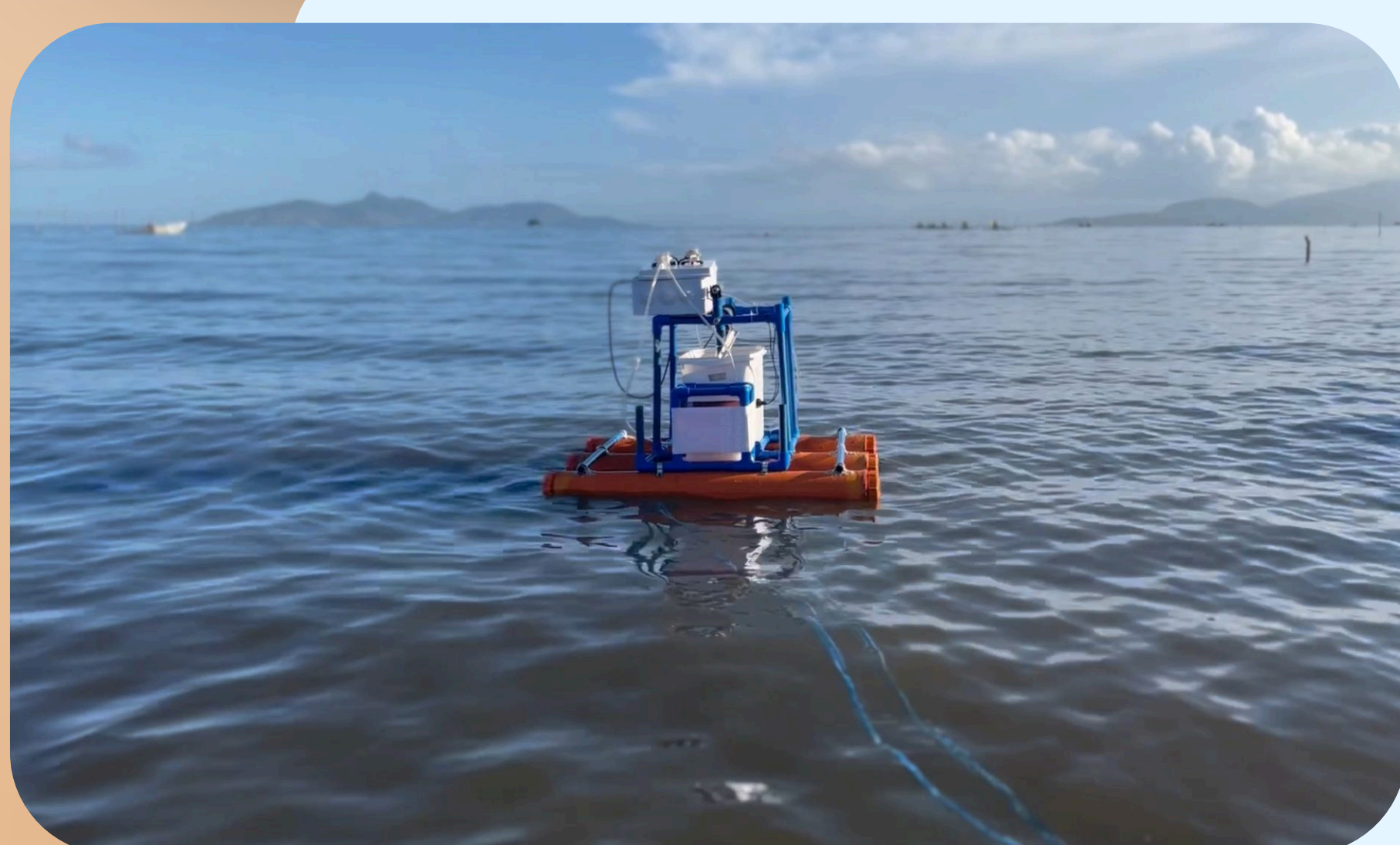
Precision

Oil detection has achieved approximately 90% precision and remains stable across various environments.

Oil recovery has reached up to 92% efficiency with a response time of less than 3 seconds.

92%

Oil Recovery



- Watercraft remained stable and navigable across diverse water conditions.
- Demonstrated consistent buoyancy, control, and reliable detection.
- Minor detection drops were observed under harsh lighting and in the presence of debris.

CONCLUSION



SANAOil is an autonomous system that integrates machine learning-based vision with mechanical recovery to address nearshore oil spills. It showed reliable navigation, accurate detection, and efficient oil extraction, making it a scalable and practical solution for environmental response.

